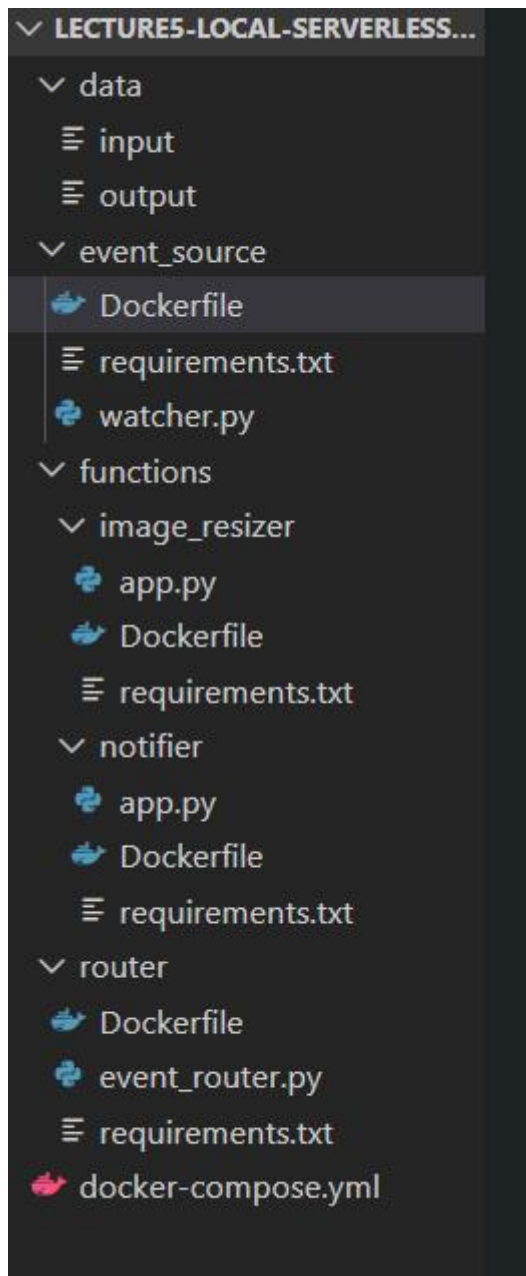


LAB5

Name: Batool wesam

Group: SE2

1-the Project Structure



2-Built and started all services

```
C:\Users\hp\Downloads\lecture5-local-serverless-lab>docker compose up --build
time="2026-06-03T12:28:52+03:00" level=warning msg="C:\Users\hp\Downloads\lecture5-local-serverless-lab\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
#1 [internal] load local base definitions
#1 reading from stdin 2.37kB done
#1 DONE 0.0s

#2 [image_resizer internal] load build definition from Dockerfile
#2 transferring dockerfile: 217B 0.0s done
#2 DONE 0.1s

#3 [event_source internal] load build definition from Dockerfile
#3 transferring dockerfile: 208B 0.0s done
#3 DONE 0.1s

#4 [router internal] load build definition from Dockerfile
#4 transferring dockerfile: 218B 0.0s done
#4 DONE 0.1s

#5 [notifier internal] load build definition from Dockerfile
#5 transferring dockerfile: 215B 0.0s done
#5 DONE 0.1s

#6 [notifier internal] load metadata for docker.io/library/python:3.12-slim
#6 DONE 4.7s

#7 [notifier internal] load .dockerignore
#7 transferring context: 2B done
#7 DONE 0.1s

#8 [router internal] load .dockerignore
#8 transferring context: 2B done
#8 DONE 0.1s

#9 [event_source internal] load .dockerignore
#9 transferring context: 2B done
#9 DONE 0.1s

#10 [image_resizer internal] load .dockerignore
#10 transferring context: 2B done
#10 DONE 0.2s

#11 [image_resizer 1/5] FROM docker.io/library/python:3.12-slim@sha256:090ba77e2958f6af52a5341f788b50b032dd4ca28377d2893dcf1ecbdfdf203
#11 resolve docker.io/library/python:3.12-slim@sha256:090ba77e2958f6af52a5341f788b50b032dd4ca28377d2893dcf1ecbdfdf203 0.1s done
#11 ...

#12 [image_resizer internal] load build context
#12 transferring context: 1.56kB 0.0s done
#12 DONE 0.1s

#13 [event_source 1/5] FROM docker.io/library/python:3.12-slim@sha256:090ba77e2958f6af52a5341f788b50b032dd4ca28377d2893dcf1ecbdfdf203
#13 resolve docker.io/library/python:3.12-slim@sha256:090ba77e2958f6af52a5341f788b50b032dd4ca28377d2893dcf1ecbdfdf203 0.1s done
#13 ...

#13 [event_source internal] load build context
#13 transferring context: 1.65kB 0.0s done
```

```
C:\Users\hp\Downloads\lecture5-local-serverless-lab>docker compose ps
time="2026-06-03T13:37:53+03:00" level=warning msg="C:\Users\hp\Downloads\lecture5-local-serverless-lab\docker-comp
ose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                                COMMAND                                SERVICE            CREATED
STATUS
event_source        lecture5-local-serverless-lab-event_source  "python watcher.py"                  event_source       13 seconds ago
Up 11 seconds
image_resizer       lecture5-local-serverless-lab-image_resizer  "python app.py"                      image_resizer      13 seconds ago
Up 12 seconds      0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp
notifier            lecture5-local-serverless-lab-notifier      "python app.py"                      notifier           13 seconds ago
Up 12 seconds      0.0.0.0:5003->5000/tcp, [::]:5003->5000/tcp
redis               redis:7-alpine                             "docker-entrypoint.s..."          redis             13 seconds ago
Up 12 seconds      0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
router              lecture5-local-serverless-lab-router        "python event_router..."          router            13 seconds ago
Up 11 seconds
```

```
C:\Users\hp\Downloads\lecture5-local-serverless-lab>
```

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\hp\Downloads\lecture5-local-serverless-lab>docker compose logs event_source
time="2026-06-03T14:24:43+03:00" level=warning msg="C:\Users\hp\Downloads\lecture5-local-serverless-lab\docker-compose.yml: the
attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
event_source | Event Source started. Watching folder: /data/input
event_source | Published event: {
event_source |   "event_id": "770b44af-6984-40a1-afd6-dce2f2f47968",
event_source |   "event_type": "image.uploaded",
event_source |   "file_name": "test_image.png",
event_source |   "file_path": "/data/input/test_image.png",
event_source |   "width": 300,
event_source |   "created_at": "2026-06-03T11:24:14.340528"
event_source | }
```

```
C:\Users\hp\Downloads\lecture5-local-serverless-lab>docker compose logs router
time="2026-06-03T14:24:59+03:00" level=warning msg="C:\Users\hp\Downloads\lecture5-local-serverless-lab\docker-compose.yml: the
attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
router | Event Router started. Waiting for events...
router | Received event: image.uploaded
router | {
router |   "event_id": "770b44af-6984-40a1-afd6-dce2f2f47968",
router |   "event_type": "image.uploaded",
router |   "file_name": "test_image.png",
router |   "file_path": "/data/input/test_image.png",
router |   "width": 300,
router |   "created_at": "2026-06-03T11:24:14.340528"
router | }
router | Error sending event to http://image-resizer:5000/resize: HTTPConnectionPool(host='image-resizer', port=5000): Max retries e
xceeded with url: /resize (Caused by NameResolutionError("HTTPConnection(host='image-resizer', port=5000): Failed to resolve 'image-r
esizer' ([Errno -5] No address associated with hostname)))
router | Sent to http://notifier:5000/notify
```

```

What's next:
  Debug this Compose error with Gordon → docker ai "help me fix this compose error"

C:\Users\hp\Downloads\lecture5-local-serverless-lab>docker compose logs image_resizer
time="2026-06-03T14:51:17+03:00" level=warning msg="C:\\Users\\hp\\Downloads\\lecture5-local-serverless-lab\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
image_resizer | * Serving Flask app 'app'
image_resizer | * Debug mode: off
image_resizer | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
image_resizer | * Running on all addresses (0.0.0.0)
image_resizer | * Running on http://127.0.0.1:5000
image_resizer | * Running on http://172.20.0.4:5000
image_resizer | Press CTRL+C to quit
image_resizer | 172.20.0.5 -- [03/Jun/2026 11:35:39] "POST /resize HTTP/1.1" 200 -

C:\Users\hp\Downloads\lecture5-local-serverless-lab>docker compose logs notifier
time="2026-06-03T14:51:34+03:00" level=warning msg="C:\\Users\\hp\\Downloads\\lecture5-local-serverless-lab\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
notifier | * Serving Flask app 'app'
notifier | * Debug mode: off
notifier | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
notifier | * Running on all addresses (0.0.0.0)
notifier | * Running on http://127.0.0.1:5000
notifier | * Running on http://172.20.0.3:5000
notifier | Press CTRL+C to quit
notifier | NOTIFICATION: {'status': 'notified', 'function': 'notifier', 'event_type': 'image.uploaded', 'file_name': 'test_image.png', 'timestamp': '2026-06-03T11:35:39.604294'}
notifier | 172.20.0.5 -- [03/Jun/2026 11:35:39] "POST /notify HTTP/1.1" 200 -

C:\Users\hp\Downloads\lecture5-local-serverless-lab>

```

3-Created a test image

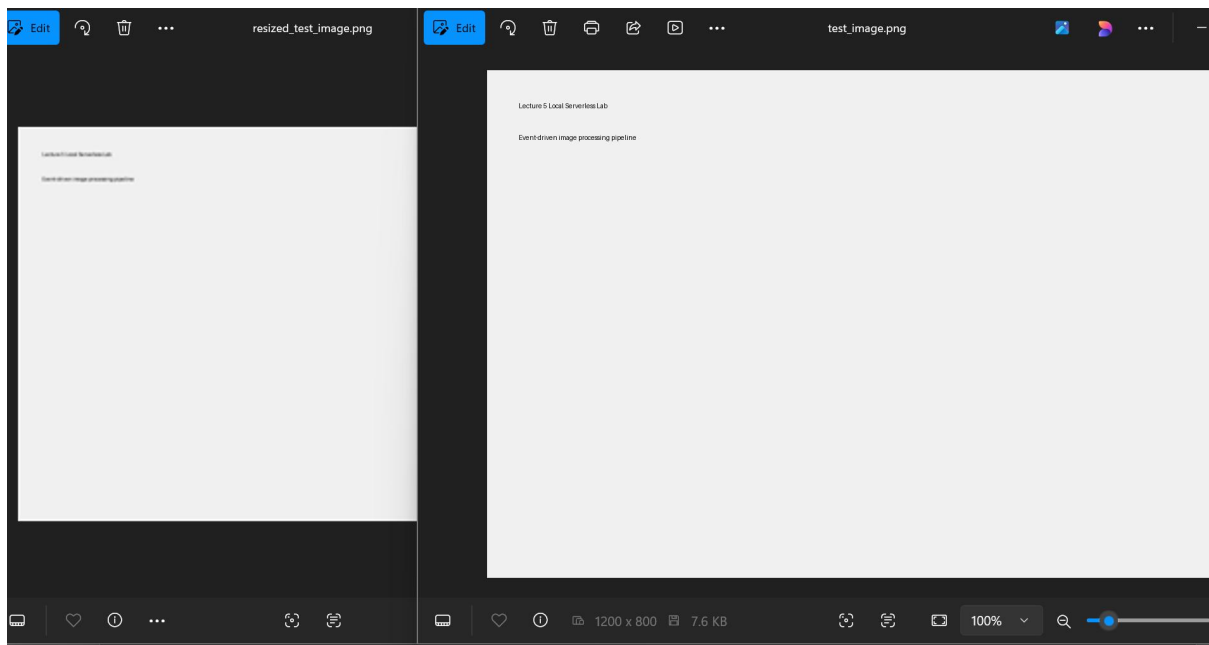
```

C:\Users\hp\Downloads\lecture5-local-serverless-lab>docker compose exec image_resizer python -c "from PIL import Image, ImageDraw; img = Image.new('RGB', (1200, 800), (240,240,240)); draw = ImageDraw.Draw(img); draw.text((50,50), 'Lecture 5 Local Serverless Lab', fill=(0,0,0)); draw.text((50,100), 'Event-driven image processing pipeline', fill=(0,0,0)); img.save('/data/input/test_image.png'); print('Created /data/input/test_image.png')"
time="2026-06-03T14:26:24+03:00" level=warning msg="C:\\Users\\hp\\Downloads\\lecture5-local-serverless-lab\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
Created /data/input/test_image.png

C:\Users\hp\Downloads\lecture5-local-serverless-lab>

```

4-Checked the output



```
C:\Windows\System32\cmd.exe
image_resizer | * Running on http://172.20.0.4:5000
image_resizer | Press CTRL+C to quit
event_source | Event Source started. Watching folder: /data/input
router | Event Router started. Waiting for events...

router | Received event: image.uploaded
event_source | Published event: {
notifier | NOTIFICATION: {'status': 'notified', 'function': 'notifier', 'event_type': 'image.uploaded', 'file_name': 'test_image.png', 'timestamp': '2026-06-03T11:3
5:39.604294'}}

image_resizer | 172.20.0.5 -- [03/Jun/2026 11:35:39] "POST /resize HTTP/1.1" 200 -
event_source | "event_id": "c188576b-56fc-4323-83a2-bff16498e3e5",

router | {

notifier | 172.20.0.5 -- [03/Jun/2026 11:35:39] "POST /notify HTTP/1.1" 200 -

View in Docker Desktop View Config Enable Watch Detach
router | "event_id": "c188576b-56fc-4323-83a2-bff16498e3e5",
event_source | "event_type": "image.uploaded",

router | "event_type": "image.uploaded",
event_source | "file_name": "test_image.png",
router | "file_name": "test_image.png",
event_source | "file_path": "/data/input/test_image.png",
router | "file_path": "/data/input/test_image.png",

event_source | "width": 300,
router | "width": 300,

event_source | "created_at": "2026-06-03T11:35:39.434279" Detach
router | "created_at": "2026-06-03T11:35:39.434279"
```

```
C:\Windows\System32\cmd.exe

router | {

notifier | 172.20.0.5 -- [03/Jun/2026 11:35:39] "POST /notify HTTP/1.1" 200 -

View in Docker Desktop View Config Enable Watch Detach
router | "event_id": "c188576b-56fc-4323-83a2-bff16498e3e5",
event_source | "event_type": "image.uploaded",

router | "event_type": "image.uploaded",
event_source | "file_name": "test_image.png",
router | "file_name": "test_image.png",
event_source | "file_path": "/data/input/test_image.png",
router | "file_path": "/data/input/test_image.png",

event_source | "width": 300,
router | "width": 300,

event_source | "created_at": "2026-06-03T11:35:39.434279" Detach
router | "created_at": "2026-06-03T11:35:39.434279"

router | }
event_source | }
router | Sent to http://image_resizer:5000/resize

View in Docker Desktop View Config Enable Watch Detach
router | Response code: 200
router | Duration: 157.89 ms
router | Response: {"function": "image-resizer", "input_file": "test_image.png", "new_size": [300, 200], "original_size": [1200, 800], "output_file": "/data/output/resized_t
est_image.png", "processing_ms": 148.52, "status": "success"}
router |
router | Sent to http://notifier:5000/notify
router | Response code: 200
router | Duration: 6.51 ms
router | Response: {'event_type': 'image.uploaded', 'file_name': 'test_image.png', 'function': 'notifier', 'status': 'notified', 'timestamp': '2026-06-03T11:35:39.604294'}
```

5-Tested the function

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\hp\Downloads\lecture5-local-serverless-lab> Invoke-RestMethod `
>> -Method Post `
>> -Uri http://localhost:5001/resize `
>> -ContentType "application/json" `
>> -Body '{"file_path":"/data/input/test_image.png","file_name":"test_image.png","width":200}'

function      : image-resizer
input_file     : test_image.png
new_size       : {200, 133}
original_size  : {1200, 800}
output_file    : /data/output/resized_test_image.png
processing_ms  : 30.25
status         : success

PS C:\Users\hp\Downloads\lecture5-local-serverless-lab>
```

6-Checked container status

☐		Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	▼	lecture5-local-s	-	-	-	0.52%	28 minutes ag	   
☐		event_source	2ec4107b3f41	lecture5-lo		0.04%	28 minutes ag	   
☐		router	d5c3a40b89f3	lecture5-lo		0%	28 minutes ag	   
☐		redis	f8d493f9abef	redis:7-alpi	6379:6379 ↗	0.46%	28 minutes ag	   
☐		notifier	22df4afb8988	lecture5-lo	5003:5000 ↗	0.01%	28 minutes ag	   
☐		image_resize	14cd2041fa8	lecture5-lo	5001:5000 ↗	0.01%	28 minutes ag	   

7-warm and cold restart

We notice that Cold request slightly slower Warm request

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\hp\Downloads\lecture5-local-serverless-lab> Invoke-RestMethod `
>> -Method Post `
>> -Uri http://localhost:5001/resize `
>> -ContentType "application/json" `
>> -Body '{"file_path":"/data/input/test_image.png","file_name":"test_image.png","width":200}'

function      : image-resizer
input_file     : test_image.png
new_size       : {200, 133}
original_size  : {1200, 800}
output_file    : /data/output/resized_test_image.png
processing_ms  : 32.33
status         : success

PS C:\Users\hp\Downloads\lecture5-local-serverless-lab> docker compose restart image_resizer
time="2026-06-03T15:05:13+03:00" level=warning msg="C:\Users\hp\Downloads\lecture5-local-serverless-lab\docker-compose.yml: the attribute 'version' is
obsolete, it will be ignored, please remove it to avoid potential confusion"
[*] restart 0/1
- Container image_resizer Restarting 1.9s
PS C:\Users\hp\Downloads\lecture5-local-serverless-lab> Invoke-RestMethod `
>> -Method Post `
>> -Uri http://localhost:5001/resize `
>> -ContentType "application/json" `
>> -Body '{"file_path":"/data/input/test_image.png","file_name":"test_image.png","width":200}'

function      : image-resizer
input_file     : test_image.png
new_size       : {200, 133}
original_size  : {1200, 800}
output_file    : /data/output/resized_test_image.png
processing_ms  : 49.13
status         : success
```

1. What is the event source in this lab?

The event source is the service that monitors the input folder (/data/input), It detects when a new image is added

2. What is the event router?

The event router is the central service that receives events from Redis, It reads the event and decides where to send it next

3. What are the event destinations?

The event destinations are:

image-resizer (processes and resizes the image)

notifier (logs or sends notification that event was handled)

4. Why is this pipeline loosely coupled?

It is loosely coupled because:

Each service works independently

They communicate only through events (Redis)

One service does not directly depend on the internal code of another

This makes the system easier to scale and maintain.

5. What happened after restarting the image-resizer container?

After restarting, the container took a short time to become ready again, during that time, requests might be slower or temporarily fail

6. How is this similar to serverless cold start?

It is similar because:

The function (container) is not always active

After restart, the first request is slower

The system needs time to initialize before handling requests

7. What is missing compared with real cloud serverless platforms?

Missing features include:

Built-in monitoring and logging dashboards

High availability across regions

Security and IAM controls

8. How could this pipeline be extended to support image moderation, OCR, or ML enrichment?

We can extend it by adding new services such as:

OCR service → extracts text from images